

Assignment1: CancerMortality

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HINT: Retain all available numeric variables except those that are not useful in explaining the target. # Questions:	

- Should we remove the outliers, apply transformations, etc.in the first analysis of the variables (Variable Analysis) ?

Load the dataset and required libraries

```
rm(list=ls())
knitr::opts_chunk$set(fig.align='center', out.width='50%') # Manage output plots width and align

train <- read.csv("train.csv")
test  <- read.csv("test.csv")

library(moments)
library(car)
```

```
## Loading required package: carData
```

Define some functions to avoid repetition of same processes

```
calculate_outliers <- function(data) {
  # Check for regular outliers
  outliers <- boxplot.stats(data)$out
  num_outliers <- length(outliers)

  # Calculate the lower and upper bounds for severe outliers
  Q1 <- quantile(data, 0.25)
  Q3 <- quantile(data, 0.75)
  IQR <- Q3 - Q1

  lower_severe_bound <- Q1 - 3 * IQR
  upper_severe_bound <- Q3 + 3 * IQR

  # Identify severe outliers
  severe_outliers <- data[data < lower_severe_bound | data > upper_severe_bound]
  num_severe_outliers <- length(severe_outliers)

  # Return the results as a list
  return(list(
    num_outliers = num_outliers,
    num_severe_outliers = num_severe_outliers
  ))
}
```

Data Preparation

In this section, a descriptive analysis is conducted on each variable, a data quality report is generated, and imputation and profiling are addressed.

Univariate Descriptive Analysis (Variable Analysis)

- Original numeric variables corresponding to qualitative concepts have to be converted to factors:
- Original numeric variables corresponding to real quantitative concepts are kept as numeric but additional factors should also be created
- Exploratory Data Analysis for each variable (numeric summary and graphic support) of the train sample.

Variable *avganncount*

avganncount is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” The difference between the mean (623.2) and the median (175) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (80 and 509, respectively) are much lower than the mean of 623.2 and the maximum value of 38.150, further suggesting skewness. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 273 outliers, and 224 of these are severe outliers.

The histogram overlaid with a normal curve highlights the deviation from a normal distribution with the same mean and standard deviation. This variable has no missing values. Lastly, the skewness indicator and Shapiro test rejection of the null hypothesis also confirm the non-normal distribution of the variable.

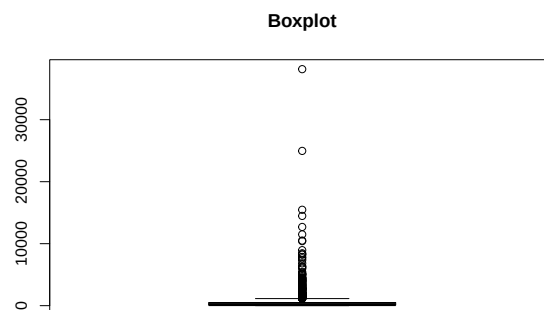
```
summary <- summary(train$avganncount); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       7.0   80.0   175.0   623.2   509.0 38150.0
```

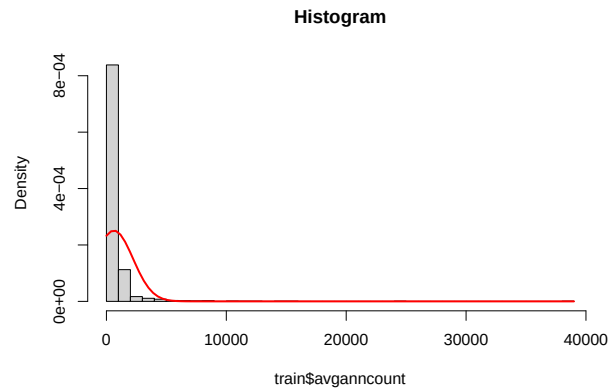
```
mm <- mean(train$avganncount)
ss <- sd(train$avganncount)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 38143
```

```
boxplot(train$avganncount, main= "Boxplot")
```



```
hist(train$avganncount,freq = F,30, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(train$avganncount)
```

```
## [1] 11.48879
```

```
shapiro.test(train$avganncount)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  train$avganncount
## W = 0.33377, p-value < 2.2e-16
```

```
calculate_outliers(train$avganncount)
```

```
## $num_outliers
## [1] 273
##
## $num_severe_outliers
## [1] 224
```

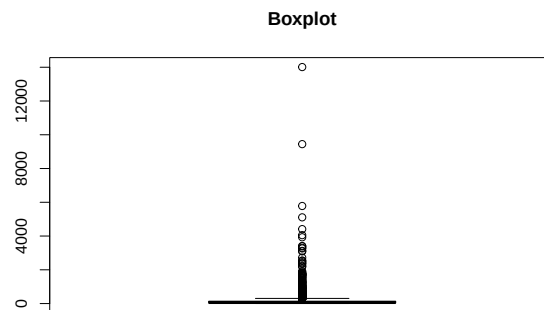
Variable *avgdeathsperyear*

In a similar way, *avgdeathsperyear* is a numerical variable expressing the “Mean number of reported mortalities due to cancer”. It has no missing values. The difference between the mean (191.6) and median (62) indicates that the distribution is not symmetric. Moreover, the first and third quartiles (29 and 140.5, respectively) are lower than the mean and the maximum value of 14010, further suggesting a skewed ditribution. With the help of graphical tools like the boxplot we can confirm the skeweness, obtaining 225 outliers (138 of which are severe), indicating a large right tail. The histogram overlaid with a normal curve highlights the deviation from a normal distribution with the same mean and standard deviation, and the Shapiro test also confirms the non-normality of the variable.

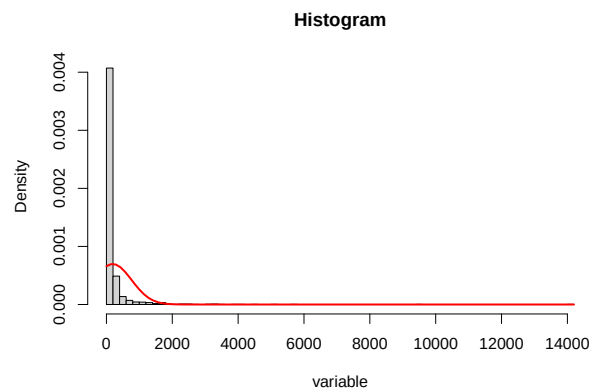
```
variable <- train$avgdeathsperyear;
summary <- summary(variable)
mm <- mean(variable)
ss <- sd(variable)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 14007
```

```
boxplot(variable, main = "Boxplot")
```



```
hist(variable,freq = F,100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(variable)
```

```
## [1] 12.4999
```

```
shapiro.test(variable)
```

```
##
## Shapiro-Wilk normality test
```

```
##
## data: variable
## W = 0.26769, p-value < 2.2e-16
```

```
calculate_outliers(variable)
```

```
## $num_outliers
## [1] 225
##
## $num_severe_outliers
## [1] 138
```

Variable *target_deathrate* :

Variable *target_deathrate* is a numerical variable that is the target of the model. It contains no missing values in the train sample. The median (178.3) and mean (178.8) values are really close, expressing the symmetry of the distribution of the variable. Thanks to the boxplot and the histogram we can graphically confirm the symmetry of the distribution.

We performed the Shapiro test to check for the normality of the data, and obtained a small p-value that rejects the null hypothesis, indicating that the variable does not follow a normal distribution. Therefore, the target variable is not normally distributed.

Moreover, it contains 35 outliers, 1 of which is a severe.

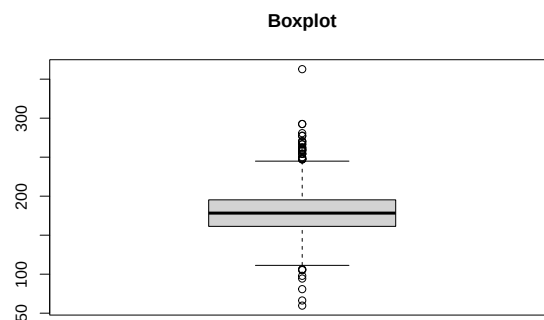
```
variable <- train$target_deathrate;
summary <- summary(variable); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      59.7  161.3   178.3   178.8   195.3   362.8
```

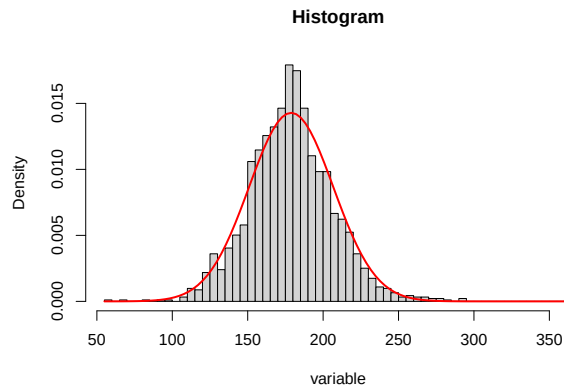
```
mm <- mean(variable)
ss <- sd(variable)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 303.1
```

```
boxplot(variable, main = "Boxplot")
```



```
hist(variable,freq = F,100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(variable)
```

```
## [1] 0.3243216
```

```
shapiro.test(variable)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  variable
## W = 0.98647, p-value = 4.149e-12
```

```
calculate_outliers(variable)
```

```
## $num_outliers
## [1] 35
##
## $num_severe_outliers
## [1] 1
```

Variable *incidencerate* :

Variable *incidencerate* is a numerical variable that expresses “Mean per capita (100,000) cancer diagnoses”. The variable has no missing values. It has similar mean (449) and median (453.5) values given its range. It contains 60 outliers, 7 of which are severe outliers. In the boxplot we can see how the most severe outliers are high and not low values, corresponding to Osceola County, Florida and Williamsburg city, Virginia. The histogram with the overlapping normal distribution shows a quite symmetrical distribution, but the Shapiro test and skewness value discard the normality of the variable’s distribution.

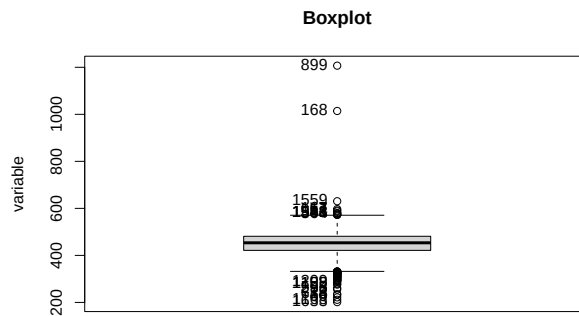
```
variable <- train$incidencerate;
summary <- summary(variable); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    201.3  421.4   453.5   449.0   481.3   1206.9
```

```
mm <- mean(variable)
ss <- sd(variable)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 1005.6
```

```
Boxplot(variable, main = "Boxplot")
```



```
## [1] 1633 1168 60 18 634 295 558 1122 1155 1209 899 168 1559 167 17
## [16] 954 1558 1548 1541 364
```

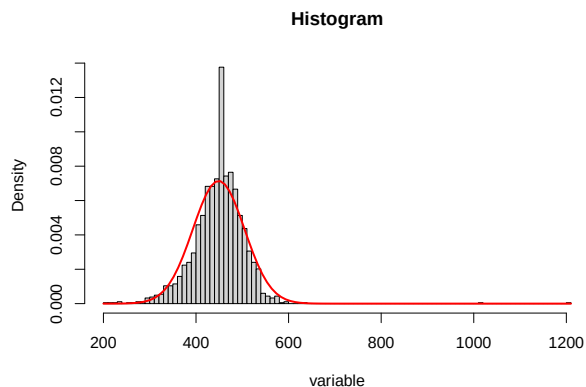
```
train[899, "geography"]
```

```
## [1] "Osceola County, Florida"
```

```
train[168, "geography"]
```

```
## [1] "Williamsburg city, Virginia"
```

```
hist(variable, freq = F, 100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss), col = "red", lwd = 2, add = T)
```



```
skewness(variable)
```

```
## [1] 1.429279
```

```
shapiro.test(variable)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: variable  
## W = 0.89577, p-value < 2.2e-16
```

```
calculate_outliers(variable)
```

```
## $num_outliers  
## [1] 60  
##  
## $num_severe_outliers  
## [1] 7
```

Variable *medincome* :

Variable *medincome* is a numerical variable that expresses the “Median income per county”. Thanks to the histogram, Shapiro test, and skewness value, we see how the distribution is neither Normal nor symmetric, but it is skewed with a long right tail. It has 69 outliers, 15 of them severe. The highest value is for Loudoun County, Virginia. There are no missing values in this variable.

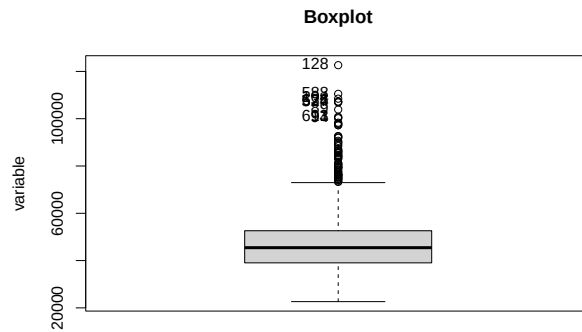
```
variable <- train$medincome;  
summary <- summary(variable); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##  22640   39031   45454   47278   52612  122641
```

```
mm <- mean(variable)  
ss <- sd(variable)  
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 100001
```

```
Boxplot(variable, main = "Boxplot")
```

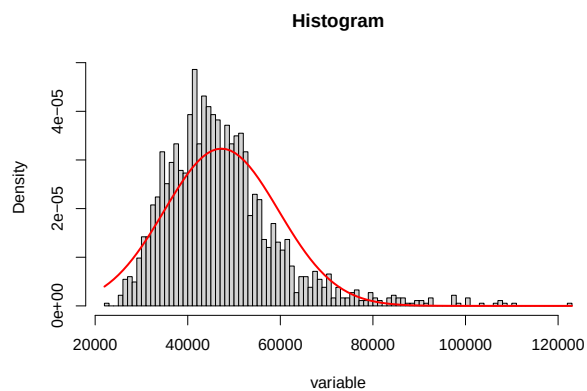


```
## [1] 128 588 104 636 574 527 89 613 91 94
```

```
train$geography[128] # Virginia
```

```
## [1] "Loudoun County, Virginia"
```

```
hist(variable,freq = F,100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(variable)
```

```
## [1] 1.466811
```

```
shapiro.test(variable)
```

```
##
## Shapiro-Wilk normality test
##
## data: variable
## W = 0.9105, p-value < 2.2e-16
```

```
calculate_outliers(variable)
```

```
## $num_outliers
## [1] 69
##
## $num_severe_outliers
## [1] 15
```

Variable *popest2015* :

Variable *popest2015* is a numerical variable that expresses the population of the county. It does not count with missing values. It has a long range and is highly skewed, with a long right tail. In fact, it contains 252 outliers, 162 of which are severe, and all of them are above the upper whisker in the boxplot. The highest value is for Los Angeles County, California. The large difference between the mean (106.841) and median (27.158), along with the spread between the 1st (12.191) and 3rd quartiles (66.880), indicates a positively skewed distribution, where extreme high values are pulling the mean upward.

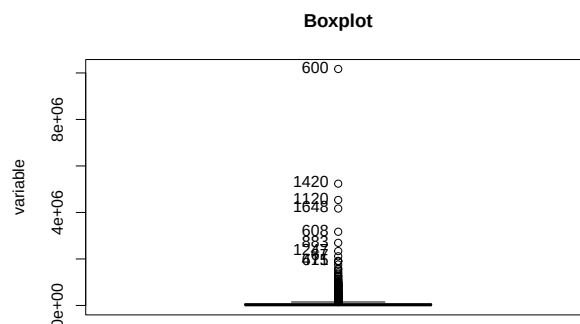
```
variable <- train$popest2015;
summary <- summary(variable); summary
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
##      829     12191    27158    106841   66880  10170292
```

```
mm <- mean(variable)
ss <- sd(variable)
range <- (summary[[6]] - summary[[1]]); range
```

```
## [1] 10169463
```

```
Boxplot(variable, main = "Boxplot")
```

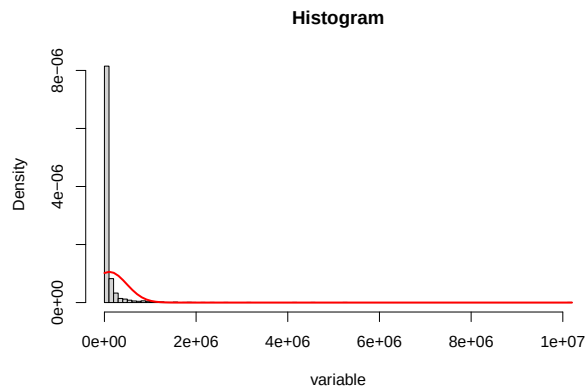


```
## [1] 600 1420 1120 1648 608 883 1247 67 615 471
```

```
train$geography[600]
```

```
## [1] "Los Angeles County, California"
```

```
hist(variable,freq = F,100, main = "Histogram")  
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(variable)
```

```
## [1] 14.70818
```

```
shapiro.test(variable)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: variable  
## W = 0.22666, p-value < 2.2e-16
```

```
calculate_outliers(variable)
```

```
## $num_outliers  
## [1] 252  
##  
## $num_severe_outliers  
## [1] 162
```

Variable *povertypersent* :

Variable *povertypersent* is a numerical variable that expresses the percent of population in poverty and has no missing values. Similarly to other variables, it has a skewed distribution with a larger right tail. It has 42 outliers, but none of them is severe.

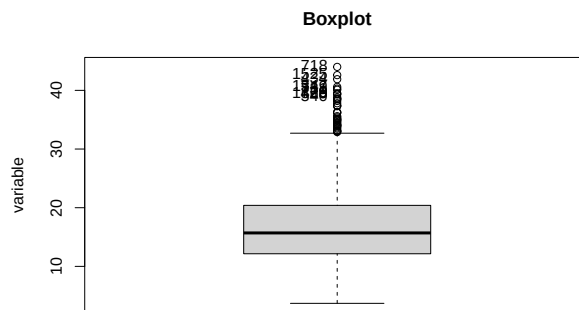
```
variable <- train$povertypercent;
summary <- summary(variable); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      3.70  12.15   15.70   16.79   20.40   44.00
```

```
mm <- mean(variable)
ss <- sd(variable)
range <- (summary[[6]] - summary[[1]]); range
```

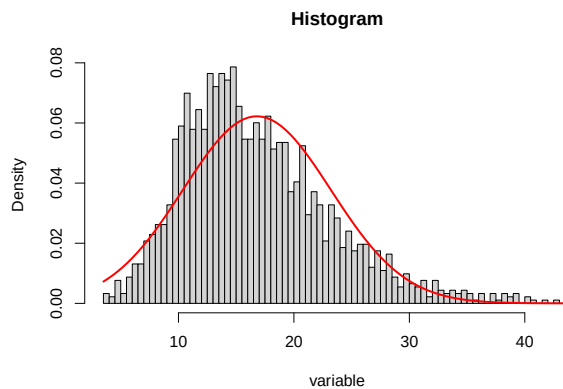
```
## [1] 40.3
```

```
Boxplot(variable, main = "Boxplot")
```



```
## [1] 718 1525 434 1547 719 731 720 1468 329 540
```

```
hist(variable,freq = F,100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
skewness(variable)
```

```
## [1] 0.8905168
```

```
shapiro.test(variable)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: variable  
## W = 0.95557, p-value < 2.2e-16
```

```
calculate_outliers(variable)
```

```
## $num_outliers  
## [1] 42  
##  
## $num_severe_outliers  
## [1] 0
```

Variable *studypercap* : I CONSIDER IT IS WORTH TO REMOVE

Variable *studypercap* is a numerical variable that expresses “Per capita number of cancer-related clinical trials per county”. It contains no missing values, and has 307 outliers (222 severe). Its distribution is non-symmetric. Note that 1162 out of 1831 observations contain value 0 in this variable, which indicates that it may only provide noise to the model. **Thus, we consider it is worth to remove the variable.**

```
variable <- train$studypercap;  
summary <- summary(variable); summary
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      0.0      0.0      0.0   148.2   76.0   9762.3
```

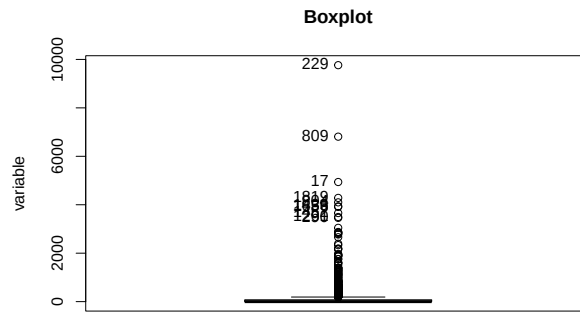
```
length(variable[variable==0])
```

```
## [1] 1162
```

```
mm <- mean(variable)  
ss <- sd(variable)  
range <- (summary[[6]] - summary[[1]]); range
```

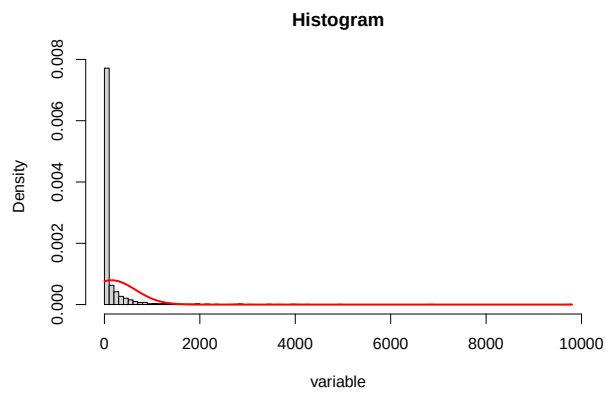
```
## [1] 9762.309
```

```
Boxplot(variable, main = "Boxplot")
```



```
## [1] 229 809 17 1819 804 1439 1656 1452 1261 290
```

```
hist(variable,freq = F,100, main = "Histogram")
curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)
```



```
calculate_outliers(variable)
```

```
## $num_outliers
## [1] 307
##
## $num_severe_outliers
## [1] 222
```

Variable *binnedi* : I CONSIDER IT IS WORTH TO REMOVE

Variable *binnedi* is a categorical variable (or factor) that expresses the median income per capita binned by decile.

```
variable <- train$binnedinc;
summary <- summary(variable); summary
```

```
## Length Class Mode
## 1831 character character
```

```
#table(variable)
#prop.table(table(variable_name))
#barplot(table(variable), main="Bar Plot of Variable", xlab="Categories", ylab="Frequency")
```

Data Quality Report

Per variable, count:

- Number of missing values.
- Number of errors (including inconsistencies).
- Number of outliers. -Rank variables according the sum of missing values (and errors).

```
# Number of missing values.
# Number of errors (including inconsistencies).
# Number of outliers.
# Rank variables according the sum of missing values (and errors).
```

Per individuals, count:

- Number of missing values
- Number of errors
- Number of outliers
- Identify individuals considered as multivariant outliers.

```
#Number of missing values
#Number of errors
#Number of outliers
#Identify individuals considered as multivariant outliers.
```

Create a variable adding the total number of missing values, outliers and errors. Describe these variables, to which other variables exist higher associations.

- Compute the correlation with all other variables. Rank these variables according the correlation
- Compute for every group of individuals the mean of missing/outliers/error values. Rank those groups according the computed mean.

```
#Include Code
```

Imputation

- Numeric Variables
- Factors

```
#Include Code
```

Profiling

- Target : Death rate

Model fitting to train sample only!!

#Include Code